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EmPOWERed 2041: Global Energy in the Next 20 Years



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According to the Intergovernmental Panel on Climate Change (IPCC), the CO₂ released till the end of 2017 had raised the average surface temperature by ~1.1°C – which can drastically affect the stability of the oceans and land mass. Globally, large economies such as China (27.2%), United States (14.6%) and India (6.8%), account for roughly half of the global CO₂ emissions. Further, as per Our World in Data's CO₂ and Greenhouse Gas Emissions database, over 73% of these GHG emissions come from global energy requirements. Hence, needless to say that clean energy generation, transmission and utilization are going to make the biggest contributions towards preventing climate change in the years to come.

In this regard, one of the major push areas worldwide in the recent past has been towards adoption of Electric Vehicles (EVs) that are powered

by electricity rather than fossil fuels. Li-ion batteries are conventionally incorporated in EVs due to their high energy density. However, force-fitting the Li-ion technology across all forms of vehicles and use cases is not only operationally challenging but also detrimental to the environment in the long run.

While Li-Ion batteries are quite viable for personal vehicles, commercial vehicles require high power capacity and longer life batteries. Supercapacitors and other high powercell based batteries can not only be rapidly charged in 10-15 minutes (compared to conventional battery charging time of 4-5 hours) but also offer a battery life of more than 15-20 years (whereas conventional battery life is around 3 years). Given that the average daily usage is significantly higher for the commercial use of vehicles, technologies like these would mean lesser number of battery packs being required over the life of the vehicle and hence lower GHG emissions per vehicle (a typical NCM111 battery produces total 73 kgCO₂ eq/kWh). Supercapacitors can also be incorporated in EVs/Hybrid vehicles to capture regenerative braking energy that not only enhances the life of the battery, but also reduces the CO₂ emissions by converting the kinetic energy of the vehicle during deceleration into electrical energy.

For Long Haul LCVs and HCVs, including trucks and buses, which are extremely difficult to be electrified using the conventional Li-Ion batteries due to limited range, Fuel Cell technology is a promising alternative which can provide the benefits of the high energy density along with a solid, stable and recyclable fuel. Also, it enables efficient and decentralized power generation that does not require any robust grid

infrastructure and makes remote rural areas clean and self-sustaining in terms of their mobility needs. Fuel Cells – whether Aluminum or Hydrogen-based – would enable utilization of excess energy generated on the grid to be used for creation of these fuels which can be used whenever required to power up the vehicles. This is extremely critical in case of intermittent energy generation from renewables such as Solar and Wind.

However, energy consumed in transportation only solves for 14% of the energy related emissions globally. Moreover, if the source for electricity generation is fossil fuels instead of renewables, then it may defeat the whole purpose of the shift. In India, 70% electricity generation is still generated from burning coal (Fig. 1).

Thus, it is important to implement renewable energy as an alternative to fossil fuels to meet the extensive global energy requirements. Wind and solar are the most abundant sources of energy available across the globe. Today, solar projects in the Middle East, China and India, or wind projects in Brazil and the US, are approaching figures lower than \$30/MWh, which is much lower than coal-based plants.

Solar power plants can be installed by industries on their rooftops, or on free land on the premises, or anywhere in the same state to generate enough power for its use. Since commercial establishments use most of their power in the daytime, solar is a good option. For a power requirement of 1 Mega Watt, companies currently pay about Rs 27 lakh as electricity bills, which can get reduced to Rs 2 lakh (the mandatory fixed rental of a grid connection) by having captive renewable generation. Government of some states like Gujarat and Rajasthan are providing incentives